

Renewable and Non-Renewable Resources in Environmental Studies (EVS)

Introduction

Natural resources are materials and substances that occur in nature and are used by humans to meet their needs. Based on their availability and regeneration capacity, resources are classified into **Renewable Resources** and **Non-Renewable Resources**.

Understanding these resources is important for environmental conservation, sustainable development, and responsible resource management.

1. Renewable Resources

Definition

Renewable resources are natural resources that can be replenished or regenerated naturally within a relatively short period of time. If used sustainably, they can be available indefinitely.

Characteristics

- Replenished naturally.
- Environment-friendly compared to fossil fuels.
- Sustainable when properly managed.
- Help reduce pollution and greenhouse gas emissions.

Examples of Renewable Resources

1. Solar Energy

- Energy obtained from the sun.
- Used in solar panels for electricity generation.
- Clean and abundant source of energy.

Example: Solar power plants in Rajasthan, India.

2. Wind Energy

- Generated using wind turbines.
- Produces electricity without air pollution.

Example: Muppandal Wind Farm, Tamil Nadu.

3. Hydropower

- Electricity generated from flowing water.
- Renewable as water is continuously recycled through the water cycle.

Example: Bhakra Nangal Dam.

4. Biomass Energy

- Produced from plant and animal waste.
- Used for cooking, heating, and electricity generation.

Example: Biogas plants using cow dung.

5. Forest Resources

- Trees and forest products can regenerate if managed properly.

Example: Sustainable forestry practices.

6. Geothermal Energy

- Energy obtained from heat inside the Earth.

Example: Geothermal power stations in Iceland.

Advantages of Renewable Resources

- Reduce dependence on fossil fuels.
- Low environmental impact.
- Unlimited supply in nature.
- Reduce greenhouse gas emissions.
- Support sustainable development.

Disadvantages

- High initial installation cost.
- Weather-dependent (solar and wind).
- Energy storage challenges.
- Land requirement for large projects.

2. Non-Renewable Resources

Definition

Non-renewable resources are resources that exist in limited quantities and cannot be replenished within a human lifetime once exhausted.

Characteristics

- Finite in nature.
- Formed over millions of years.
- Cannot be replaced quickly.
- Excessive use leads to depletion.

Examples of Non-Renewable Resources

1. Coal

- Fossil fuel formed from ancient plant remains.
- Used in thermal power plants.

Example: Coal mines in Jharkhand and Odisha.

2. Petroleum (Crude Oil)

- Used to produce petrol, diesel, and other fuels.

Example: Mumbai High Oil Field.

3. Natural Gas

- Cleaner fossil fuel compared to coal.
- Used in industries and households.

Example: Krishna-Godavari Basin.

4. Nuclear Fuels

- Uranium and thorium used in nuclear reactors.

Example: Uranium mines in Jharkhand.

5. Metallic Minerals

- Iron, copper, gold, and aluminum.

Example: Iron ore mines in Karnataka.

Advantages of Non-Renewable Resources

- High energy output.
- Reliable and continuous energy supply.
- Existing infrastructure supports their use.

- Easier storage and transportation.

Disadvantages

- Limited availability.
- Cause air, water, and soil pollution.
- Contribute to global warming.
- Extraction damages ecosystems.
- Resource depletion concerns.